



Part I

Lab Assignment-9 (Introduction to Object-Oriented Programming)

Department: CSE
(22CPP209)

Subject: OOAD Lab

Objective: Understanding *Polymorphism and Abstract classes*.

Question 1. Extend the package-delivery system problem to incorporate polymorphism and abstract classes by modifying the base class `Package` to be an abstract class by making its `calculateCost()` function pure virtual. This will enforce that each derived class implements its own version of `calculateCost()`. Creates **objects** of both `TwoDayPackage` and `OvernightPackage`. Stores pointers to these objects in a **vector of `Package*`** (base class pointers). Iterates through the vector, calling the `calculateCost()` function polymorphically to compute the cost for each package. For each package, output its identifying information (e.g., sender and recipient) and the calculated cost.

Question 2: Create three unrelated classes: **Electronics**, **Clothing**, and **FoodItem**. Each class should have unique attributes such as brand and warranty for Electronics, size and fabric type for Clothing, and expiration date for FoodItem. Define an abstract class called **ShoppingItem** with a pure virtual function `calculateDiscount()` that calculates the discount for each item based on its category. Implement the function for each class to return the appropriate discount percentage or amount.

Write a program that creates objects for each class, places them in a vector of `ShoppingItem` pointers, and iterates through the vector to display each item's information and calculated discount.